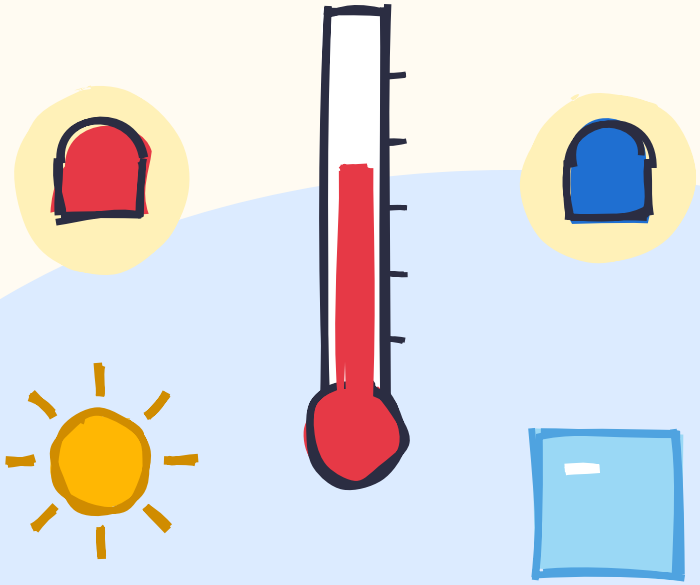
Little Explorers · Class 6

Hot & Cold Temperature Light

Red means warm. Blue means cool!

Temperature sensor | Red and blue lights | Choosing a number



Did you know?
Your body stays at about 37 degrees, even on a cold winter day!

Let's wonder...

Can our hands tell the truth?

A tricky experiment

Put one hand in **cold** water and one hand in **warm** water. Wait.
Now put **both** hands in the middle bowl.
One hand says **warm**! The other says **cold**!



Our hands are confused!

Which hand is right?

Use a sensor!

Today: we build a machine that measures hot and cold with **numbers**, and shows it with a red or blue light.

LEARN

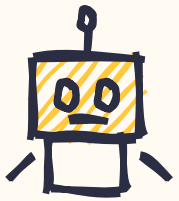
Learning aims

Today I will learn to...

1. I can say if something is **hot** or **cold** using a **number**.
2. I can make a **red** or **blue** light switch on.
3. I can **choose the number** where red turns on.

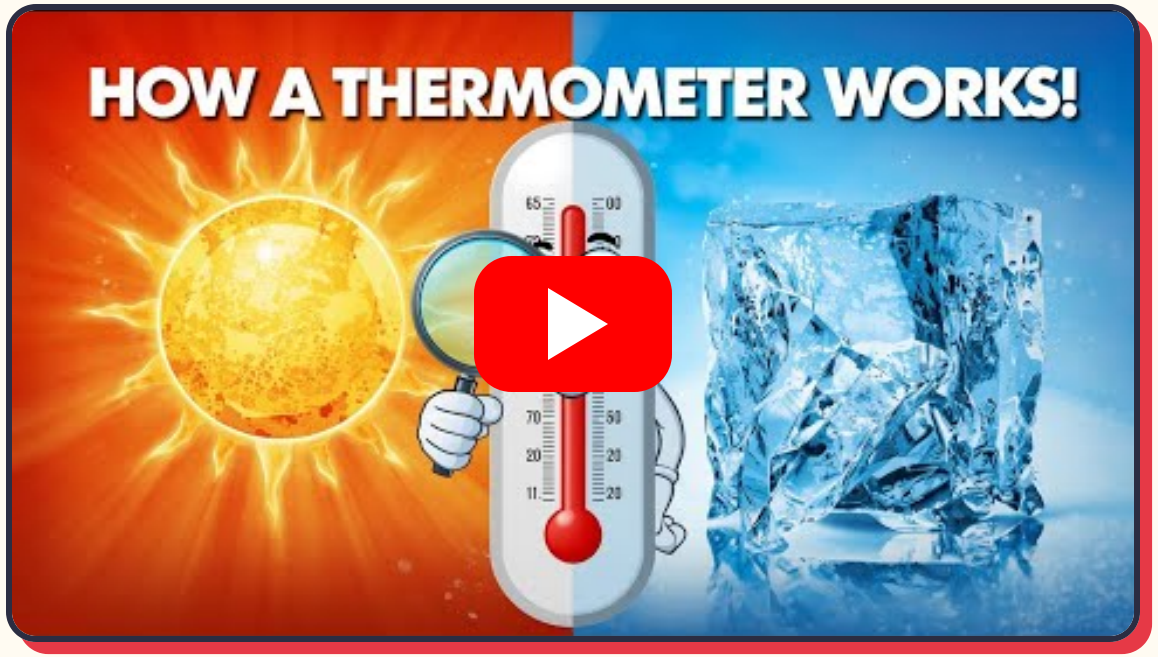
Keywords

hot · cold · warm · cool · temperature · degrees · sensor



Watch and think

Video time!



How a Thermometer Works

Whimsy Wonders · about 5 minutes · youtu.be/XYEX_2oxj9g

Watch for...

1. What does a **thermometer** measure?
2. What happens to the line when it gets **hot**?
3. Where do we use thermometers?

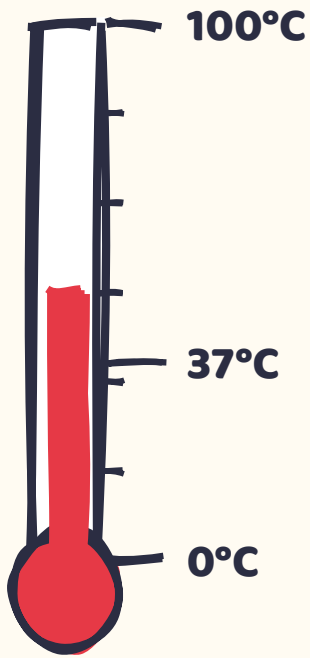
Scan to watch

After the video: Show with your arm: arm up = hot, arm down = cold. Your partner says a place and you show it!

LEARN

Big idea

Temperature is a NUMBER



100°C boiling water (very hot!)

37°C inside our body

0°C ice

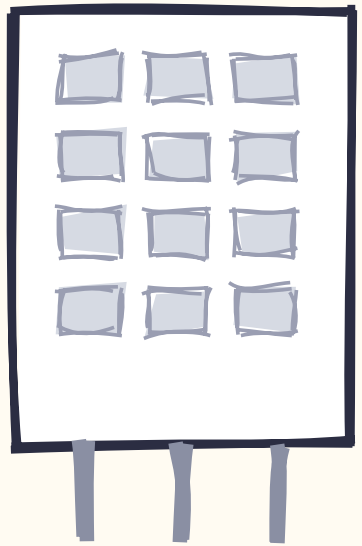
Degrees
We write °C and say degrees. Bigger number = hotter!

Try it
Is 30 hotter or colder than 20? What about 5?

LEARN

Meet our helper

The temperature sensor



A tiny thermometer

This white box feels the air around it. It tells the Arduino the temperature as a **number**, like **27**.

Fun fact

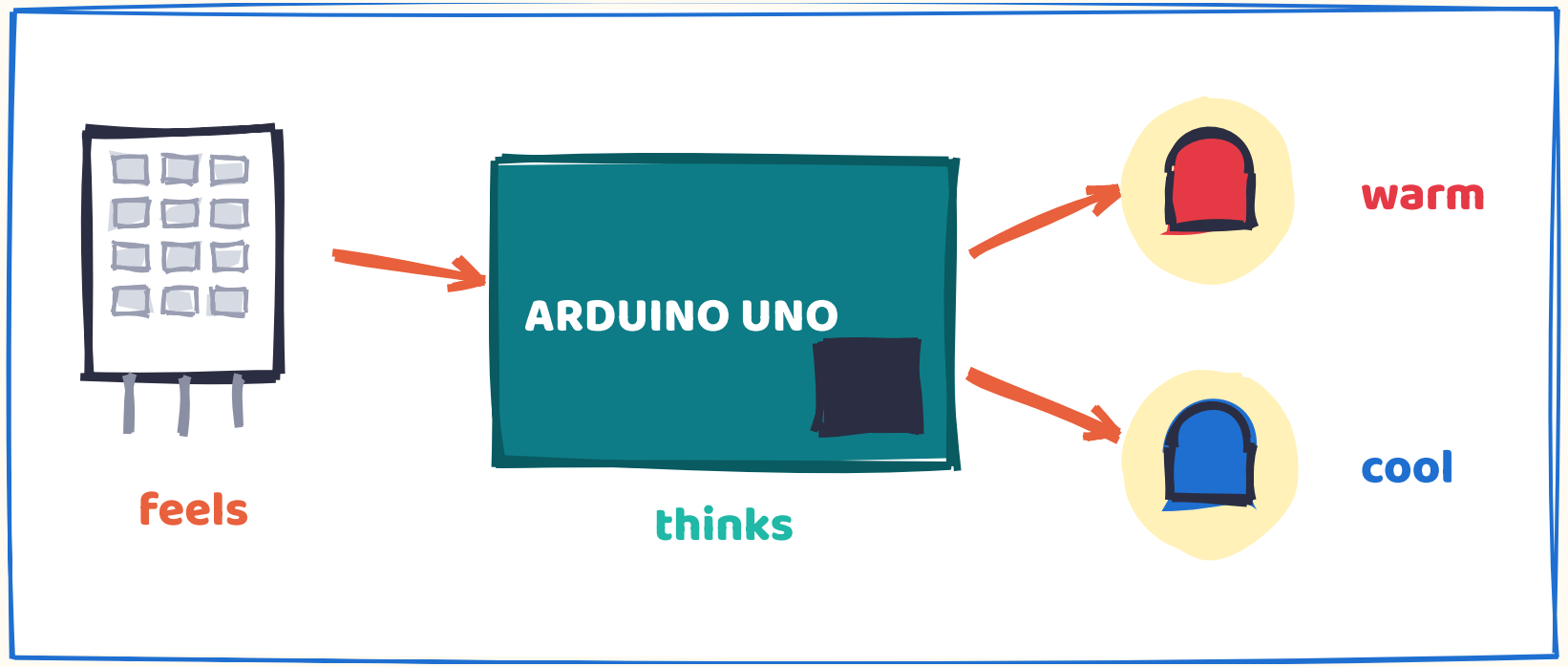
It can also tell if the air is damp (humidity). Like after rain!

Keep it dry!

It is not waterproof. Never put it in water or in your mouth.

Our machine

Feel, think, show



How it works

The sensor **feels** the temperature.
The Arduino **thinks**: is it bigger than our number?
The lights **show** the answer.

Teacher: the sensor and two LEDs are pre-wired (data pin 2, red LED pin 12, blue LED pin 11). Children plug in, code and test.

LEARN

Explorers!

Where is the warmest place?

Place	Number	Light
Near the window	_____ °C	red / blue
Under the fan	_____ °C	red / blue
Near the door	_____ °C	red / blue
Inside a school bag	_____ °C	red / blue

Talk about it

Which place was **warmest**? Which was **coolest**? Why do you think so?

Teacher

Carry the laptop with the Arduino on a tray. Walk slowly together!

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Hot things can hurt!
2. Thermometers keep us safe
3. Why do we check temperature?
4. Hot or cold?
5. Let's code our light
6. Make it change colour
7. Choose YOUR hot number

Stay safe

Hot things can hurt!

- **Never touch** a stove, kettle, iron or hot pan.
- **Never put** our sensor in water, tea or your mouth.
- **Ask a grown-up** to check if food or drinks are too hot.
- **Cool a burn** with cool running water and tell a grown-up straight away.



Remember

Our sensor measures the AIR. It is not for checking a fever.

Real heroes

Thermometers keep us safe



Doctors

Check if you have a fever. 38°C or more means tell a grown-up!



Fridges

Keep food cold so germs do not grow



Farmers

Protect plants from frost and heat



If you could put a temperature light anywhere at home, **where** would you put it?

Draw it!

Draw your invention and show the class.

Hot and cold

Why do we check temperature?



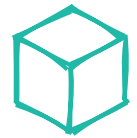
Fever
Is my little brother poorly? Check with a thermometer



Weather
Should I wear a jumper or a T-shirt today?



Cooking
Is the milk too hot to drink?



The fridge
Keeps food cold so it stays fresh

? Where do YOU feel hot or cold? Kitchen? Car in summer? Under a fan?

Ask at home
Do you have a thermometer? Where is it kept?

MAKE

Class game

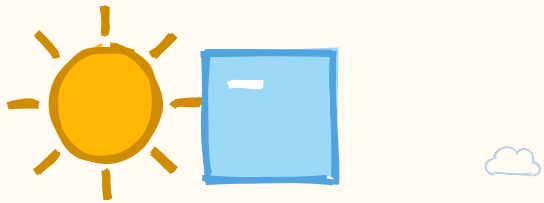
Hot or cold?

How to play

1. The teacher shows a picture card.
2. **HOT**? Stand up tall like the sun!
3. **COLD**? Curl up small like an ice cube!
4. Cards: ice cream, cup of tea, snowman, fire, the fridge, summer sun

Then discuss

Is a cup of tea **always** hot? What happens if you wait one hour?



Big idea

Hot things cool down and cold things warm up over time.

MAKE

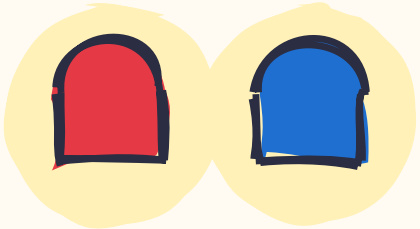
Time to build!

Let's code our light

```
when Arduino starts
forever
    set temp to temperature (pin 2)
    if temp > 28
        turn ON red light
        turn OFF blue light
    else
        turn ON blue light
        turn OFF red light
    wait 2 seconds
```

Read it like a story

Read the temperature.
IF it is more than 28: **red**.
ELSE: **blue**.
Wait, then check again!

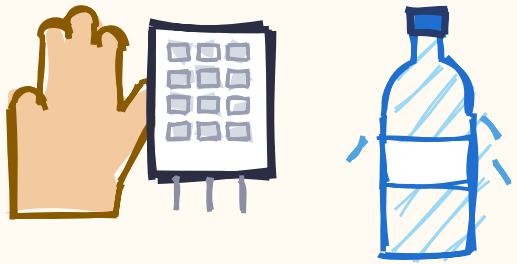


Why wait 2 seconds?
Our sensor is a bit slow. It needs a little rest between readings.

Experiment!

Make it change colour

Try this	I think…	Number	Light
Leave it on the table	red or blue?	_____	_____
Hold it gently in both hands	red or blue?	_____	_____
Hold it near a cold bottle	red or blue?	_____	_____
Blow on it	red or blue?	_____	_____



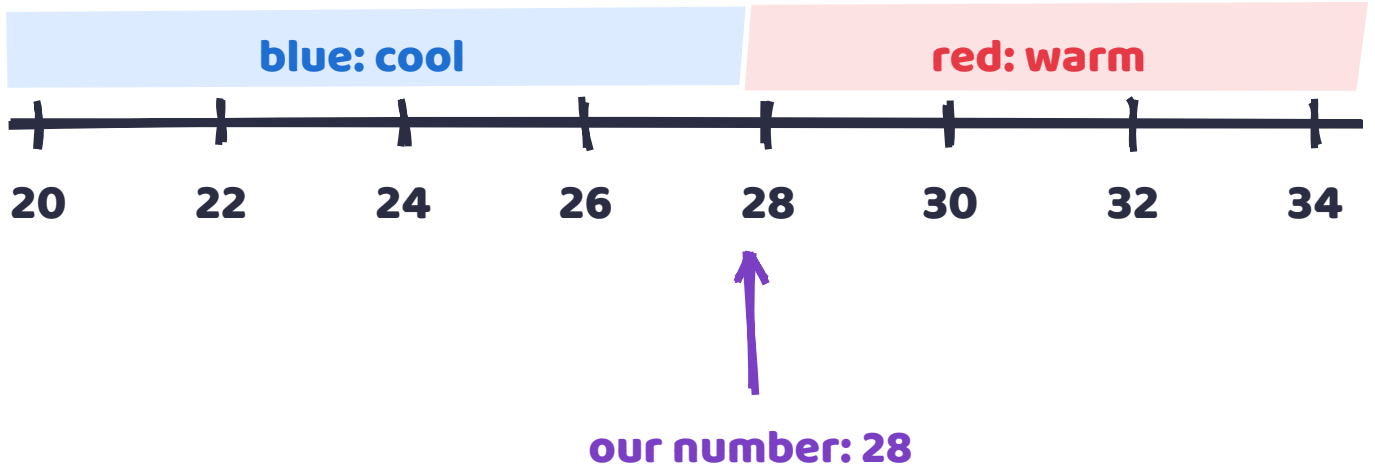
Be patient! The number changes **slowly**. Count to 20 before you look.

Cold bottle tip
Wrap the bottle in a plastic bag so no water drips on the sensor.

MAKE

You decide!

Choose YOUR hot number



Your turn

Change **28** in the code to a new number. Try **25**. Try **32**. When does the light change easily? When does it never go red?

Bigger or smaller?

If the room is 27 and your number is 25, which light shows?

Engineer word

The number where things change is called a THRESHOLD.

MAKE

Level up!

Hot and cold challenges

Choose your challenge



Just right: add a green light for a middle number, not too hot and not too cold.



Beep when hot: make the buzzer beep when the red light comes on.



Blink fast: make the red light blink when it is VERY hot.



Class chart: write everyone's hot number on the board. Whose was the biggest?

Think like an engineer

Why do you think a fridge has a light that switches on when you open the door?

Boss level

Make the light change at TWO numbers: blue, green, then red!

Show what you know

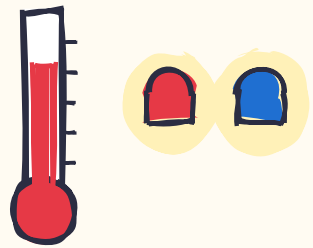
What did we learn?

- [x] Temperature is a number in degrees (°C)
- [x] Bigger number = hotter
- [x] Our sensor is a tiny thermometer for the Arduino
- [x] IF it is hot: red. ELSE: blue
- [x] We chose our own hot number (threshold)

Exit ticket: is 35°C hotter or colder than 20°C?

At home

With a grown-up, look at a weather app. What is today's temperature?



Next class: a Plant Water Alarm!